

Lecture Notes

Course: Complex Analysis

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Topic: Complex Integral

Theorem 8.23 (Darboux Inequality)

Proof. Write the integral in polar form and arrive to

$$R = \left| \int_a^b \gamma \right| = e^{-i\phi} \int_a^b \gamma = \int_a^b e^{-i\phi} \gamma$$

and by taking real parts one can deduce the inequality

$$\Re(R) = R = \Re \left(\int_a^b e^{-i\phi} \gamma \right) = \int_a^b \Re \left(e^{-i\phi} \gamma \right) \leq \int_a^b |e^{-i\phi} \gamma| = \int_a^b |\gamma|$$

and the inequality preserving property of the integral follows directly from the fact that if $f(I) \leq g(I)$ and if $\mathcal{P}^*$ is a tagged partition of I , then

$$S(f; \mathcal{P}^*) \leq S(g; \mathcal{P}^*).$$

The desired inequality now follows directly from

$$\left| \int_{\gamma} f \right| = \left| \int_a^b (f \circ \gamma) \gamma' \right| \leq \int_a^b |f \circ \gamma| |\gamma'| \leq M \int_a^b |\gamma'| = ML(\gamma).$$

Further, if γ is a contour and not just a smooth curve, then by the triangle inequality

$$\left| \int_{\gamma} f \right| = \left| \sum_i \int_{\gamma_i} f \right| \leq \sum_i \left| \int_{\gamma_i} f \right| \leq \sum_i \int_a^b |(f \circ \gamma_i) \gamma'_i| \leq M \sum_i \int_a^b |\gamma'_i| = ML(\gamma)$$

Theorem 8.24 (Properties of the Complex Integral)

Proof. Linearity follows directly from the linearity of the HK-integral, that is,

$$\int_{\gamma} (\lambda f + g) = \int_a^b ((\lambda f + g) \circ \gamma) \gamma' = \lambda \int_a^b (f \circ \gamma) \gamma' + \int_a^b (g \circ \gamma) \gamma' = \lambda \int_{\gamma} f + \int_{\gamma} g.$$

For the opposite contour we recall that $-\gamma(t) = \gamma(a + b - t)$ and thus letting $r(t) = a + b - t$ we have $-\gamma = \gamma \circ r$ so that by the chain rule

$$\int_{-\gamma} f = \int_a^b (f \circ -\gamma) (-\gamma)' = - \int_a^b (f \circ \gamma \circ r) (\gamma' \circ r)$$

so that using $u = r$ we get $du = -dt$ and hence

$$\int_{-\gamma} f = - \int_b^a -(f \circ \gamma) \gamma' = \int_b^a (f \circ \gamma) \gamma' = - \int_a^b \gamma (f \circ \gamma) \gamma' = - \int_{\gamma} f.$$

Theorem 8.25 (Reparametrisations preserve the track)

Proof. Let σ, γ be curves with σ a reparametrisation of γ , i.e., there exists a bijection of $[\alpha, \beta]$ onto $[a, b]$ such that $\sigma = \gamma \circ \phi$, then

$$\sigma([\alpha, \beta]) = \gamma(\phi([\alpha, \beta])) = \gamma([a, b])$$

where the last equality follows from the fact that ϕ is a bijection. More generally, if Σ is a reparametrisation of Γ , then for each curve Γ_i composing Γ , Σ_i is a reparametrisation thereof, so

$$\Sigma_i(J_i) = \Gamma_i(I_i) \quad (\text{no sum})$$

and hence $\Sigma(\bigcup_i J_i) = \Gamma(\bigcup_i I_i)$ since $\Gamma = \Gamma_i \chi_{I_i}$, and $\Sigma = \Sigma_i \chi_{J_i}$.

Theorem 8.26 (Reparametrisations preserve integrals)

Proof. Suppose γ is a curve and let σ be a reparametrisation thereof. We compute

$$\int_{\sigma} f = \int_{\alpha}^{\beta} dt f(\sigma(t)) \sigma'(t) = \int_{\alpha}^{\beta} dt f(\gamma(\phi(t))) \gamma'(\phi(t)) \phi'(t),$$

thus on letting $u = \phi$ we get $du = \phi' dt$

$$\int_{\phi(\alpha)}^{\phi(\beta)} du f(\gamma(u)) \gamma'(u) = \int_a^b (f \circ \gamma) \gamma' = \int_{\gamma} f.$$

The proof for the general case (contours) is now a direct consequence of the proven result and the additivity theorem.

Theorem 8.27 (The Fundamental Theorem of Contour Integration (First Form))

Proof. a) $\Rightarrow$ b) If f has a primitive on Ω , and we take any closed contour γ on Ω , then taking F to be the primitive, we can see that $(F \circ \gamma)' = (f \circ \gamma) \gamma'$ and hence by the Fundamental Theorem of Calculus

$$\int_{\gamma} f = F(\gamma(b)) - F(\gamma(a)),$$

which vanishes since, by assumption, $\gamma(a) = \gamma(b)$.

b) $\Rightarrow$ c) Let $\Gamma = \{\gamma_i\}_{i=1}^n, \Sigma = \{\sigma_i\}_{i=1}^m$ be two contours in Ω joining z to w , then define the reparametrisation of the paths so that

$$\tilde{\gamma}_i : I_i \rightarrow \Gamma^{\dagger} \quad \text{and} \quad \tilde{\sigma}_i : J_i \rightarrow \Sigma^{\dagger}, \quad I_i = \left[\frac{i-1}{2n}, \frac{i}{2n} \right], \quad J_i = \left[\frac{i-1}{2m} + \frac{1}{2}, \frac{i}{2m} + \frac{1}{2} \right],$$

denote $\tilde{\Gamma} = \{\tilde{\gamma}_i\}$ and $\tilde{\Sigma} = \{\tilde{\sigma}_i\}$. Define the contour $\Upsilon : [0, 1] \rightarrow \Upsilon^{\dagger}$ as

$$\Upsilon(t) = \begin{cases} \gamma_i & t \in I_i \\ -\sigma_i & t \in J_i \end{cases},$$

and notice that Υ simply takes z to w for $t \in [0, 1/2]$ and then w to z for $t \in [1/2, 1]$ so that Υ is a closed contour, and thus the hypothesis plus the additivity theorem yields

$$0 = \int_{\Upsilon} f = \int_{\tilde{\Gamma}} f + \int_{-\tilde{\Sigma}} f \implies \int_{\tilde{\Gamma}} f = \int_{\tilde{\Sigma}} f \implies \int_{\Gamma} f = \int_{\Sigma} f,$$

where the last implication is due to Theorem 8.26.

c) $\Rightarrow$ a) Suppose the integral is independent of the path and thus using the insight of the Fundamental Theorem for a real variable we define γ to be any path in Ω connecting $z_0 \in \Omega$ to z and propose

$$F(z) = \int_{\gamma} f$$

to be the primitive for f on Ω . In order to show this we compute the difference

$$F(z+h) - F(z) = \int_{\Xi} f,$$

where Ξ is any path connecting z to $z + h$. Next, we estimate a bound for the difference between the differential quotient and $f(z)$,

$$\left| \frac{F(z+h) - F(z)}{h} - f(z) \right| = \left| \frac{1}{h} \int_{\Xi} dw f(w) - f(z) \right| = \frac{1}{|h|} \left| \int_{\Xi} dw f(w) - f(z)h \right|, \quad (*)$$

using the fact that Ω is open we can conclude that there exists $r > 0$ such that $B_r(z) \subset \Omega$, and since the election of path is irrelevant, choose the line segment joining z to $z + h$ (we will choose h near enough to z to ensure that h is still in $B_r(z)$ and thus this election is permitted) so that

$$\int_{\Xi} dw = h,$$

and thus $(*)$ becomes

$$\frac{1}{|h|} \left| \int_{\Xi} dw f(w) - f(z) \int_{\Xi} dw \right| = \frac{1}{|h|} \left| \int_{\Xi} dw (f(w) - f(z)) \right|,$$

we now make an estimate on the difference $f(w) - f(z)$ which is achieved by the fact that f is continuous, meaning that the difference can be made less than ε for any given $\varepsilon > 0$ provided that $|z - w| < \delta_\varepsilon$ for some $\delta_\varepsilon > 0$ (the elements w in Ξ are in the line segment joining z to $z + h$, i.e., of the form $w = z + th$ so it is enough to choose $|h| < \delta_\varepsilon$ because $0 \leq t \leq 1$), then by the Darboux inequality we have that the bound for $(*)$ is

$$\frac{1}{|h|} \sup_{w \in \Xi} |f(w) - f(z)| L_{\Xi} < \frac{1}{|h|} \varepsilon |h| = \varepsilon,$$

with the length of Ξ being $|h|$ and the supremum of the difference being ε . We therefore conclude that given any $\varepsilon > 0$, we can choose $\delta = \inf\{\delta_\varepsilon, r\}$ so that $|h| < \delta$ implies

$$\left| \frac{F(z+h) - F(z)}{h} - f(z) \right| < \varepsilon,$$

that is, $F'(z) = f(z)$ for any $z \in \Omega$.

Theorem 8.41 (Cauchy Integral Formula)

Proof. For any closed contour Γ , its interior is open, meaning that for any $z_0 \in \text{Int } \Gamma$ there exists a number $R > 0$ such that $B_R(z_0) \subseteq \Omega$, this way, $f(z)/(z - z_0)$ is in $H(\Omega \setminus \overline{B_R}(z_0))$ and letting $C_r = B_r(z_0)$ with $0 < r \leq R$ we get (due to Cauchy's theorem for multiply connected regions)

$$\oint_C dz \frac{f(z)}{z - z_0} = \oint_{C_r} dz \frac{f(z)}{z - z_0},$$

so that

$$\oint_{C_r} dz \frac{f(z)}{z - z_0} = \oint_{C_r} dz \frac{f(z) + f(z_0) - f(z_0)}{z - z_0} = \oint_{C_r} dz \frac{f(z) - f(z_0)}{z - z_0} + \oint_{C_r} dz \frac{f(z_0)}{z - z_0}$$

the second integral is already known to be $2\pi i f(z_0)$ and for the first one notice that since f is continuous, given any $\varepsilon > 0$ we can choose $\delta > 0$ such that $|f(z) - f(z_0)| < \varepsilon/2\pi$ provided that $|z - z_0| < \delta$ and for $z \in B_r(z_0)$ we have $|z - z_0| = r$ meaning that $\varepsilon/(2\pi r)$ is a bound for the integrand, so for $r < \delta$ we have that

$$\left| \frac{f(z) - f(z_0)}{z - z_0} \right| < \frac{\varepsilon}{2\pi r}$$

and thus Darboux's inequality now implies (clearly, $L_{B_r(z_0)} = 2\pi r$)

$$\left| \oint_{C_r} dz \frac{f(z) - f(z_0)}{z - z_0} \right| < \frac{\varepsilon}{2\pi r} 2\pi r = \varepsilon,$$

whence the conclusion

$$\lim_{r \rightarrow 0} \oint_{C_r} dz \frac{f(z) - f(z_0)}{z - z_0} = 0$$

is immediate, and thus by the previous remarks

$$\oint_C dz \frac{f(z)}{z - z_0} = 2\pi i f(z_0),$$

and the proof is done.

Lemma 8.43 (Leibniz rule lemma)

Proof. We let $z \in B_{R/2}(z_0)$ and define

$$f(z) = \frac{1}{2\pi i} \int_{B_R(z_0)} d\zeta \frac{f(\zeta)}{\zeta - z}$$

and thus

$$\frac{f(z) - f(z_0)}{z - z_0} = \frac{1}{2\pi i} \int_{B_R(z_0)} d\zeta \frac{f(\zeta)}{(\zeta - z)(\zeta - z_0)}. \quad (*)$$

We now see that for $\zeta \in B_R(z_0)$ we have $|\zeta - z| \geq |\zeta - z_0| - |z - z_0| = R - |z - z_0|$ but since $z \in B_{R/2}(z_0)$, $|z - z_0| < R/2$ from which $-|z - z_0| > -R/2$ and hence $|\zeta - z| \geq R - R/2 = R/2$ and so $\frac{1}{|\zeta - z|} < \frac{2}{R}$. Thus

$$\begin{aligned} \left| \frac{f(\zeta)}{(\zeta - z)(\zeta - z_0)} - \frac{f(\zeta)}{(\zeta - z_0)^2} \right| &= \left| \frac{f(\zeta)}{(\zeta - z_0)} \left(\frac{1}{\zeta - z} - \frac{1}{\zeta - z_0} \right) \right| = \left| \frac{f(\zeta)(z - z_0)}{(\zeta - z_0)^2(\zeta - z)} \right| \\ &\leq M \frac{2}{R} \frac{|z - z_0|}{R^2} \end{aligned} \quad (**)$$

let us now consider the difference

$$\left| \frac{f(z) - f(z_0)}{z - z_0} - \frac{1}{2\pi i} \int_{B_R(z_0)} d\zeta \frac{f(\zeta)}{(\zeta - z_0)^2} \right|,$$

which in view of (*) and (**) as well as the Darboux inequality, yields

$$\left| \frac{1}{2\pi i} \int_{B_R(z_0)} d\zeta \left(\frac{f(\zeta)}{(\zeta - z)(\zeta - z_0)} - \frac{f(\zeta)}{(\zeta - z_0)^2} \right) \right| \leq \frac{1}{|2\pi i|} M \frac{2|z - z_0|}{R^3} 2\pi R = \frac{2M}{R^2} |z - z_0|,$$

and the conclusion is immediate.

Theorem 8.44 (Leibniz Rule)

Proof. For any fixed $z_0 \in \Omega$ we let $R > 0$ be such that $\overline{B_R(z_0)} \subset \Omega$. Let¹

$$M = \sup_{z \in B_R(z_0), \zeta \in \gamma} |\phi(z, \zeta)|.$$

By hypothesis $\phi(z, \zeta)$ is holomorphic in Ω and thus for $z \in B_{R/2}(z_0)$ and $\zeta \in \gamma^\dagger$ we may apply the previous lemma to get

$$\left| \frac{\phi(z, \zeta) - \phi(z_0, \zeta)}{z - z_0} - \partial_z \phi(z_0, \zeta) \right| \leq 2M \frac{|z - z_0|}{R^2}.$$

Considering the integral on the LHS we can deduce that

$$\left| \frac{g(z) - g(z_0)}{z - z_0} - \int_\gamma d\zeta \partial_z \phi(z_0, \zeta) \right| = \left| \int_\gamma d\zeta \left[\frac{\phi(z, \zeta) - \phi(z_0, \zeta)}{z - z_0} - \partial_z \phi(z_0, \zeta) \right] \right|$$

¹Continuous functions attain a maximum on compacts

which by the previous lemma and the Darboux inequality can be bounded by $L(\gamma)2M|z - z_0|/R^2$ and now the result follows immediately.

Remark (Cauchy-Riemann completion)

Proof. The converse follows from using that

$$f^{(k)}(z_0) = \partial_x^k u(x_0, y_0) + i \partial_x^k v(x_0, y_0) = \partial_y^k v(x_0, y_0) - i \partial_y^k u(x_0, y_0).$$

Theorem 8.52 (Liouville's Theorem.)

Proof. Cauchy's inequality and f being entire implies

$$|f'(z)| \leq \frac{M}{r}.$$

Theorem 8.53 (Fundamental Theorem of Algebra)

Proof. Let $p(z)$ be a nonconstant polynomial, i.e., a polynomial of degree $n \geq 1$ and suppose $p(z)$ has no roots. Then the function

$$g(z) = \frac{1}{p(z)}, \quad \lim_{z \rightarrow \infty} \frac{1}{p(z)} = 0$$

is entire and bounded since its limit at infinity converges, so that it is constant by Liouville's theorem, clearly, a contradiction.

Theorem 8.55 (Taylor's Theorem)

Proof. by Cauchy's integral formula, we get

$$f(z) = \frac{1}{2\pi i} \oint_{B_\delta(z_0)} d\zeta \frac{f(\zeta)}{\zeta - z}$$

for any $z \in B_\delta(z_0)$, $\delta < R$. By stroke of insight we see that

$$\frac{1}{\zeta - z} = \frac{1}{\zeta - z_0} \left(\frac{1}{1 - \frac{z - z_0}{\zeta - z_0}} \right) = \frac{1}{\zeta - z_0} \left(\frac{1 - \left(\frac{z - z_0}{\zeta - z_0}\right)^n}{1 - \frac{z - z_0}{\zeta - z_0}} + \frac{\left(\frac{z - z_0}{\zeta - z_0}\right)^n}{1 - \frac{z - z_0}{\zeta - z_0}} \right),$$

and then

$$f(z) = \frac{1}{2\pi i} \oint_{B_\delta(z_0)} d\zeta \frac{f(\zeta)}{\zeta - z} = \sum_{k=0}^{n-1} \frac{1}{2\pi i} \oint_{B_\delta(z_0)} \frac{f(\zeta)}{(\zeta - z_0)^{k+1}} (z - z_0)^k + R_n,$$

with

$$R_n = \frac{1}{2\pi i} \oint_{B_\delta(z_0)} d\zeta \left(\frac{z - z_0}{\zeta - z_0} \right)^n \frac{f(\zeta)}{\zeta - z},$$

and since

$$|R_n| \leq \frac{M\delta}{\delta - R} \left(\frac{R}{\delta} \right)^n,$$

then $R_n \rightarrow 0$ as $n \rightarrow \infty$.